

3. (8pts) Consider the sequence recursively defined by $b_1 = 3, b_2 = 4$, and

$$\underline{b_{n+1} = 2b_n + b_{n-1}}$$

for $n \geq 2$. Show that $\gcd(b_n, b_{n+1}) = 1$ for all $n \geq 1$.

$$3, 4, 2 \cdot 4 + 3, 2 \cdot 1 + 4, \dots$$

$$3, 4, 11, 26, \dots$$

no common factors other than 1.

Proof by induction: We proceed by induction. $P(n)$: " $\gcd(b_n, b_{n+1}) = 1$ ".

Base's Step: $P(1)$ = " $\gcd(b_1, b_2) = \gcd(3, 4) = 1$ ", which is true.

Inductive Step: Assume $k \geq 1$ is arbitrary.

Assume $P(1), \dots, P(k)$.

We will show $P(k+1)$, i.e., $\gcd(b_{k+1}, b_{k+2}) = 1$.

By defn:

$$b_{k+2} = 2 \cdot b_{k+1} + b_k \quad (1)$$

$$\cancel{b_{k+1} = 2 \cdot b_k + b_{k-1} \quad (2)}$$

By $P(k)$,

$$\gcd(b_k, b_{k+1}) = 1.$$

$$\exists u, v \in \mathbb{Z} \text{ s.t. } ub_{k+1} + vb_{k+2} = 1$$

k
 $b_{k+2}, b_{k+1},$
 b_k, s, t

So by Bezout's theorem, we can choose $P(k+1): \cancel{\gcd(b_k, b_{k+1}) = 1.}$
 $s, t \in \mathbb{Z}$ st. $sb_k + tb_{k+1} = 1$ — (3)
 (4)

~~idea~~: By (1) and (3):

$$s(b_{k+2} - 2b_{k+1}) + tb_{k+1} = 1$$

$$\Rightarrow sb_{k+2} - 2sb_{k+1} + tb_{k+1} = 1$$

$$\Rightarrow \underbrace{sb_{k+2}} + \underbrace{(t-2s)b_{k+1}} = 1$$

$\therefore$ By Bezout's theorem, $\gcd(b_{k+1}, b_{k+2}) = 1$, as desired. $\square$

Bezout:

$$\gcd(a, b) =$$

$$\min \{ sa + tb : sa + tb \geq 1, s, t \in \mathbb{Z} \}$$

$$\gcd(a, b) = 1 \Leftrightarrow \exists s, t \in \mathbb{Z} (sa + tb = 1).$$

Alternate :

Observe :

$$b_1 = 3$$

$$b_2 = 4,$$

$$\boxed{\underline{b_{n+1}} = \underline{2b_n + b_{n-1}}.}$$

$$a = b_{n+1}$$

$$d = b_n$$

$$q = 2$$

$$r = b_{n-1}$$

~~DA~~

Given $a, d > 0 \exists$ unique

$$q, r \in \{0, \dots, d-1\}$$

$$\text{s.t. } a = dq + r$$

b_n are positive integers.

Also, $b_n > b_{n-1} \forall n \geq 2$.

$$\begin{aligned} \text{By DA+EA: } \gcd(b_{n+1}, b_n) \\ = \gcd(b_n, b_{n-1}). \end{aligned}$$

~~EA~~

$$\text{If } u = dq + r$$

$$\gcd(a, d) = \gcd(d, r)$$

Graph Th: Show that if $G = (V, E)$ is a connected graph with at least two vertices, there exists a vertex $x \in V$ such that $G' = G - \{x\}$ is connected.



($\forall G$ (G connected $\wedge |V| \geq 2 \rightarrow \exists x \in V$ s.t.

$G' = (V - \{x\}, E - \{e : e \text{ incident to } x \text{ in } G\})$ is connected)

G has 1 connected comp.

G contains a spanning tree T

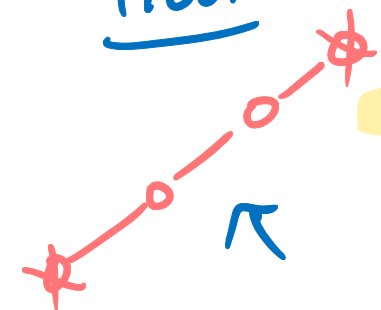


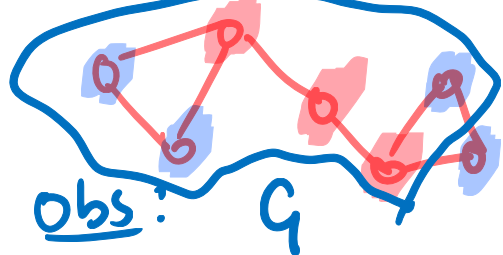
G is connected if $\forall x, y \in V$ ($\exists$ a path in G between x and y).

Proof: Assume $G = (V, E)$ is a connected graph with $|V| \geq 2$.

Choose a spanning tree $T = (V, F) \subseteq G$.

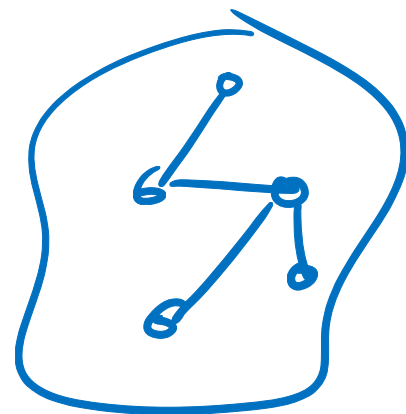
Let x be a leaf of T (because every tree





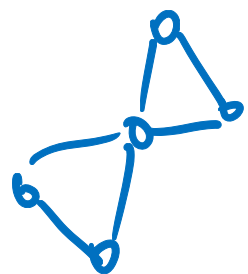
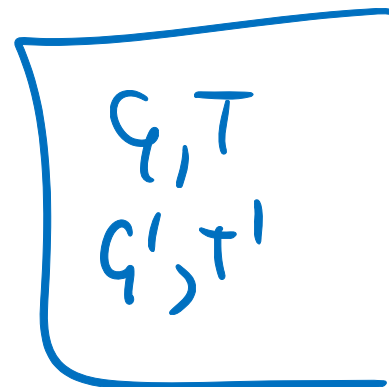
with $|V| \geq 2$ has a leaf).

Obs: G
 If there is a vertex
 with degree 1, deleting
 it produces a connected
 graph.

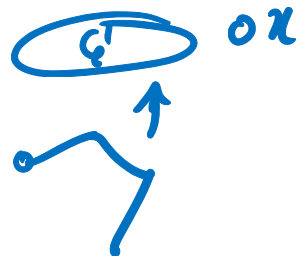
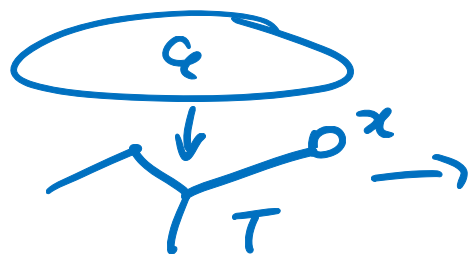


$$\text{Let } G' = G - \{x\}.$$

$$T' = T - \{x\}.$$



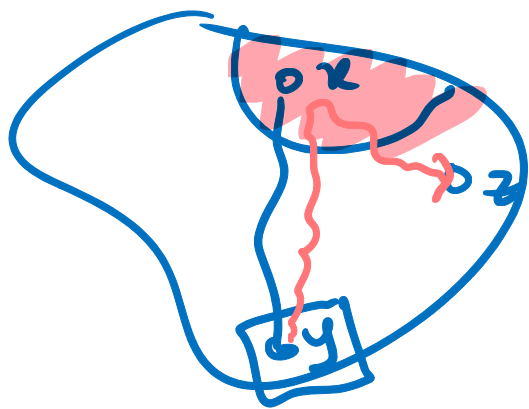
Notice that T' is a tree
 and T' is a spanning subgraph of G' .
 $\therefore T'$ is a Spanning tree of G' , so G' is
 connected, as desired. $\square$



Idea: Assume G arbitrary connected, $|V| \geq 2$.

Let y be an arbitrary vertex of G .

(Choose x s.t. $\text{dist}(x, y) \geq \text{dist}(z, x) \forall z \in G$.)



Let $G' = G - \{x\}$.

We will show G' is connected.

We will show: $\forall z \neq x$, there is
a path between y and z
in G' .

Assume not. Then $\exists z$ s.t. every path π in G

from y to z , π visits x . This
implies $\text{dist}(y, z) > \text{dist}(y, x)$, contradicting
our choice of x . D

Recap:

Graph $G=(V,E)$

$E =$ Set or multiset of pairs of vertices
simple multigraph

$\deg(v) =$ # of edges incident with v .

$$\sum_{v \in V} \deg(v) = 2|E|.$$

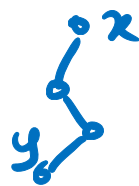
Subgraph $H=(W,F)$ of $G=(V,E)$

if $W \subseteq V, F \subseteq E$

Spanning if $W=V$.

Connected Component: Maximal connected subgraph; Can partition any $G = G_1 \cup G_2 \cup \dots \cup G_k$.

Spanning tree



path between x, y

$$\pi = x=x_0, e_1, x_1, \dots, e_m, x_m=y$$

Circuit: $x_0=x_m$.



Cycle: Circuit, no edges or vertices repeated except $x_0=x_m$.

→ can also view as a subgraph, forgetting order.

Connected: $\forall x \neq y \in V (\exists \text{ path } \pi \text{ between } x \text{ and } y)$.

2-coloring
↔
no odd ckt.

k-coloring: $f: V \rightarrow \{1, \dots, k\}$ s.t.
 $\forall xy \in E (f(x) \neq f(y))$

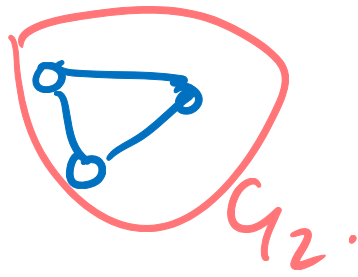
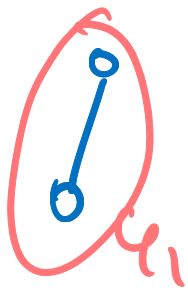
Tree: Connected acyclic graph.
(leaves, $n-1$ edges, unique path)

Euler circuit: Circuit which uses every edge once.

$k=D+1$
Coloring

even
degree

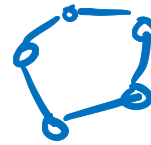
eg.



Examples :

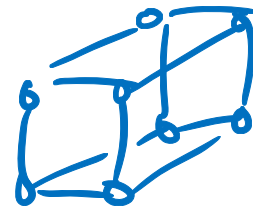
- x K_n complete graph (All edges present)

- x Cycle graph



- x Trees

- x hypercube



Common Arguments :

- x If T is a tree, x is a leaf, $T - \{x\}$ is a tree.

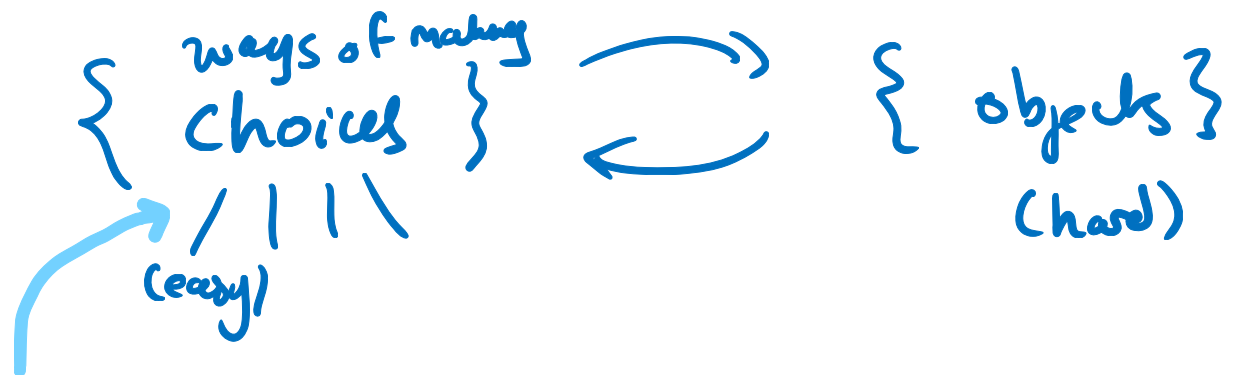
- x To prove something by induction, choose a vertex x or edge e and consider $G - \{x\}$ or $G - \{e\}$.

- x To prove about arbitrary G , consider connected components $G = G_1 \cup G_2 \cup \dots \cup G_k$.

x If C is a cycle, e is an edge,
 $C - \{e\}$ is connected.

Counting:

To count a collection of objects, define
a process for uniquely specifying
an object in the collection.



Rules: Division, Sum, Product.

Pigeonhole: What are the pigeons & holes?

Foundations

Logic — proposition
— quantifiers
— proof — clear
— correct
— proof techniques.

Sets

Content (from scratch, unfamiliar)

Number Th

Graph Th

Counting

Probability